

Question

An element is named after a celestial body in the solar system, and the density of its simple substance exhibits an anomalous increase compared to its two adjacent elements. Now, by reacting a trifluoromethanesulfonate of a certain oxidation state of this element with potassium tris(3,5-dimethyl-1-pyrazolyl)hydroborate ($KTp^{3Me,5Me}$) under anhydrous and oxygen-free conditions, a mixed-ligand eight-coordinate complex was obtained. Please predict the ligands and metal center contained in this complex.

- A.** U(III) with the $Tp^{3Me,5Me}$ ligand and the 3,5-dimethylpyrazol-1-ide ligand
- B.** Pu(IV), featuring the $Tp^{3Me,5Me}$ ligand and bis(3,5-dimethyl-1-pyrazolyl)dihydroborate ($Bp^{3Me,5Me}$) ligand
- C.** Ce(III) with the ligand $Tp^{3Me,5Me}$ and the 1,1,3,3-tetrakis(3,5-dimethyl-1-pyrazolyl)diboroxane-1,3-diide ligand
- D.** U(III), containing the $Tp^{3Me,5Me}$ ligand and the 1,1,3,3-tetrakis(3,5-dimethyl-1-pyrazolyl)diboroxane-1,3-diide ligand
- E.** Ce(III) with the $Tp^{3Me,5Me}$ ligand and 3,5-dimethylpyrazol-1-ide ligand
- F.** All of the above options are incorrect.

Answer

Correct Answer: **E**

Detailed Explanation

Based on the information that it is named after a celestial body, it can be inferred that the element is one of He, Se, Pd, Te, Ce, U, Np, or Pu.

CHECKPOINT

1 PTS

The element is one of He, Se, Pd, Te, Ce, U, Np, or Pu

Furthermore, the abnormally increased density excludes He, Se, Pd, and Te. For U, Np, and Pu, since f-electrons can participate in metallic bonding, the density increases sequentially from Ac to Np, reaching its maximum at Np, and then decreases at Pu. Thus, the element is also not U, Np, or Pu.

For Ce, since it can use four valence electrons to participate in metallic bonding, its density is nearly equivalent to that of Pr, which uses only three valence electrons for bonding. Therefore, the element is Ce.

CHECKPOINT

2 PTS

The element is Ce

Regarding the formation of complexes, since the conditions are strictly anhydrous and oxygen-free, the $Tp^{3Me,5Me}$ ligand will not hydrolyze to produce oxygen-containing ligands but will instead undergo B-N bond cleavage to form a pyrazolyl anion ligand. Thus, the correct answer is E.

CHECKPOINT

1 PTS

$Tp^{3Me,5Me}$ ligand does not hydrolyze to produce oxygen-containing ligands

CHECKPOINT

1 PTS

The ligand is $Tp^{3Me,5Me}$ and 3,5-dimethylpyrazolyl anion